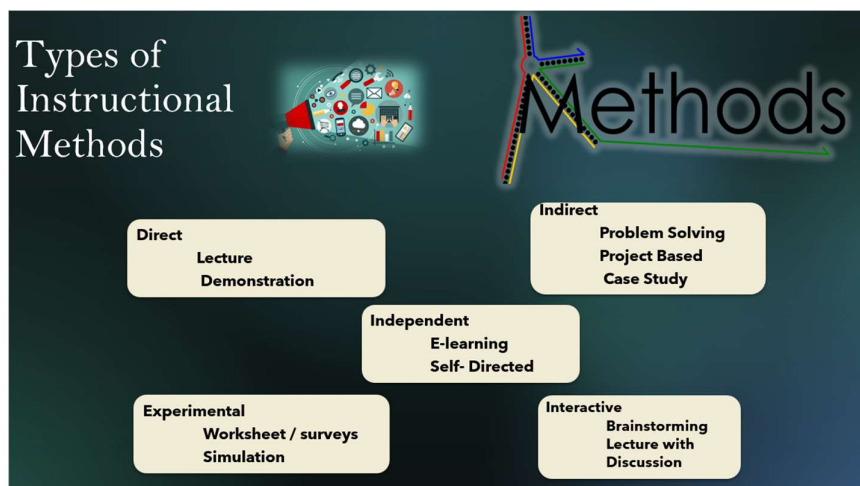


Outcome Based Teaching Learning Process

2.1 Process Followed for Teaching – Learning Process

- Adherence to Academic calendar
- Course plan
 - Lesson Plan
- Classroom Environment
- Teaching Aids
- Instructional Strategies
 - Direct, Indirect, Independent, Interactive, Experimental
- Workshops and Seminar
- Online Training programmes
- Learning Management System
- Maintenance of Course file
- Professional Development Programmes
- Through MOOCs
- Offline Training programmes
- Smart Classroom
- Lecture Capture Systems
- Student mentoring System
- Instructional Materials

Instructional Methods



Problem Solving Method

- Use of the scientific method
- Students play an active role in determining and seeking the critical information needed to solve the problem.
- Problems often have multiple solutions and relate to the real world

Problem Solving Method - Process

- Identify the Problem
- Define the Problem
- Explore possible Solutions
- Evaluate and Select solution
- Implement and follow-up the solution

2.2. Quality of Student Capstone Project

A Capstone Project is a culminating academic activity in the final year that demonstrates a student's ability to:

- Apply engineering knowledge
- Solve complex engineering problems
- Integrate technical, professional, ethical, and societal considerations

NBA evaluates capstone projects based on depth, relevance, and outcomes, not merely completion.

Key Expectations:

- Addresses complex engineering problems
- Demonstrates design, analysis, or innovation
- Involves open-ended problem solving
- Reflects industry/societal relevance
- Shows evidence of lifelong learning and teamwork

Mapping Capstone Project with SDG

Step-1: Identify Societal Problem

- Water scarcity, energy efficiency, healthcare access, waste disposal, etc.

Step-2: Link Technical Solution to SDG Target

- Clearly mention **SDG number and target** in project proposal.

Step-3: Show Engineering Contribution

- Design, analysis, modeling, implementation, and validation.

Step-4: Assess Impact

- Quantitative or qualitative impact on sustainability or society

“Final year capstone projects are aligned with UN Sustainable Development Goals to inculcate societal responsibility and sustainability. Each project explicitly maps its objectives to relevant SDGs and NBA Graduate Attributes (Programme Outcome). The assessment reflects students' ability to design sustainable engineering solutions addressing real-world challenges.”

2.3 Internship/Industrial Training

Purpose and Relevance

Internship and Industrial Training are integral components of the curriculum designed to:

- Bridge the gap between theory and practice
- Expose students to real-world engineering problems
- Develop professional skills, ethics, and lifelong learning
- Strengthen attainment of Program Outcomes (POs) and Graduate Attributes (GAs)

These activities strongly support Complex Engineering Problem Solving and Continuous Quality Improvement (CQI).

Internship / Industrial Training Structure

(a) Pre-Internship Phase

- Industry identification and MoU
- Student orientation
- Allocation of faculty mentor
- Definition of learning objectives
- Mapping with POs / PSOs / SDGs (where applicable)

(b) During Internship / Training

- Perform assigned technical tasks
- Follow industrial safety and ethical practices
- Use modern engineering tools
- Maintain daily work diary / logbook
- Interact with industry professionals

(c) Post-Internship Phase

- Submission of internship report
- Presentation / viva voce
- Assessment using rubrics
- Feedback from industry mentor
- Reflection on learning outcomes

“The department facilitates internships and industrial training to enhance students’ practical exposure and employability. These activities are well-structured, monitored by faculty mentors, and assessed using defined rubrics mapped to POs and PSOs. Industry feedback and student reflections indicate significant improvement in technical competence, professional ethics, and problem-solving abilities. Continuous improvements are implemented based on feedback analysis.”

2.4. Seminar and Mini/Micro Projects

Seminars and Mini/Micro Projects are integral curricular components designed to:

- Develop research orientation and self-learning
- Strengthen problem analysis and technical communication
- Introduce design thinking and innovation at early stages
- Prepare students for capstone projects and industry exposure

These activities significantly contribute to achieving Program Outcomes (POs) and Graduate Attributes (GAs), particularly GA2, GA3, GA4, GA9, and GA11.

Mini/Micro Projects provide students with hands-on experience in solving well-defined engineering problems, emphasizing design, implementation, and validation at an early stage of the program.

Outcome Attainment & CQI

Outcome Measurement

- Rubric-based evaluation
- CO–PO mapping and attainment
- Student performance analysis

Continuous Improvement Examples

- Moving from toy problems → real-world problems
- Increased use of Engineering tools
- Interdisciplinary mini projects
- Integration with internships & capstone projects

“The department offers Seminar and Mini/Micro Project courses to inculcate research orientation, problem analysis, and design skills among students. These activities are systematically planned, monitored by faculty mentors, and assessed using rubrics mapped to Program Outcomes. The outcomes demonstrate progressive improvement in technical depth, innovation, and communication skills, supporting continuous quality improvement.”

2.5. Case Studies and Real-Life Examples

Purpose and Relevance

Case Studies and Real-Life Examples are systematically integrated into the curriculum to:

- Enhance problem analysis and critical thinking
- Expose students to complex engineering problems
- Bridge theory with professional practice
- Develop decision-making, ethics, and societal awareness

These activities strongly support attainment of Program Outcomes (POs) and Graduate Attributes (GAs), particularly GA2, GA3, GA6, GA8, GA9, and GA11.

Nature of Case Studies Used

Case studies are drawn from:

- Industry practices and failures
- Engineering design and operational challenges
- Safety, ethics, and sustainability issues
- Contemporary technologies and standards

They are aligned with course outcomes (COs) and program specialization.

“The department systematically integrates case studies and real-life engineering examples into core and elective courses to enhance students’ problem analysis and decision-making skills. These activities expose students to complex engineering scenarios involving technical, societal, ethical, and sustainability considerations. Assessment through rubrics mapped to Program Outcomes demonstrates improved attainment and supports continuous quality improvement.”

2.6. SWAYAM/NPTEL/MOOC/Self Learning

(Number of students registered, certification and POs/PSOs addressed.)

Prepare table with complete details and check the level of the online courses. Prepare the table for CAY, CAYm1, CAYm2 and CAYm3

2.7. Solving Complex Engineering Problems Incorporating Sustainability Goals

(Provide details of core courses (Project based learning, problem-based learning), mini projects, integrated design projects, capstone projects, hackathon or any other activity-based learning towards solving complex engineering problems targeting relevant SDGs.)

2.8. Steps Taken for Enhancing Industry Institute Partnerships

The department has taken systematic steps to strengthen Industry–Institute Partnerships (IIP) to:

- Bridge the gap between curriculum and industry requirements
- Enhance employability, practical skills, and innovation
- Promote industry-relevant teaching, learning, and assessment
- Support attainment of Program Outcomes (POs) and Graduate Attributes (GAs), especially GA3, GA5, GA8, GA9, GA10, and GA11

These initiatives contribute significantly to Complex Engineering Problem Solving and Continuous Quality Improvement (CQI).

Form the Industry Advisory Board (IAB)

- Constituted with senior industry professionals
- Periodic meetings to review:
 - Curriculum relevance
 - Emerging technology inclusion
 - Skill gaps and employability trends

- Suggestions incorporated through BoS and CQI actions

“The department has established strong Industry–Institute Partnerships through MoUs, industry advisory board inputs, internships, industry-defined projects, and expert interactions. These initiatives ensure curriculum relevance, enhance practical skills, and improve employability. Continuous feedback from industry is systematically analyzed and used for curriculum improvement and outcome attainment.”